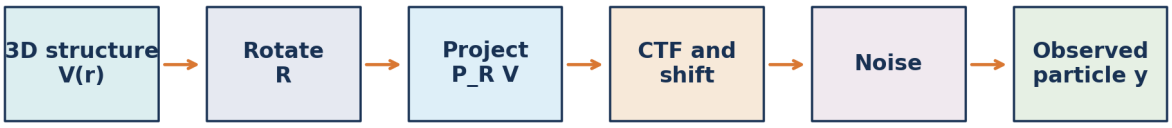
A MATHEMATICAL INTRODUCTION TO SINGLE-PARTICLE CRYO-EM

From 2D Particle Images to 3D Structures

Reconstruction, Statistical Classification and Conformational Heterogeneity

A self-contained textbook chapter with prerequisites, derivations, worked examples, experimental workflows, and reading references



$$y_i = C_i S_{t_i} P_{R_i} V + \varepsilon_i$$

Chapter II • **Companion to *Focus, Defocus, Scattering and the CTF***

Prepared 28 October 2026 | Focus: CryoSPARC, RELION, 3DVA, RECOVAR, cryoDRGN, 3DFlex

Contents

How to use this chapter **4**

1 A bird's-eye view of the reconstruction problem **4**

1.1 What data do we actually have? . . . 4

1.2 One formula for the entire imaging process . . . 5

1.3 What counts as successful reconstruction? . . . 5

2 Mathematical prerequisite I: vectors, Fourier transforms, and rotations **5**

2.1 Vectors, matrices, and inner products . . . 5

2.2 What does a Fourier transform measure? . . . 5

2.3 What is a 3D rotation? . . . 6

3 Mathematical prerequisite II: probability and inference **6**

3.1 Random variables, distributions, and expectations . . . 6

3.2 Gaussian noise and why least squares appears . . . 6

3.3 Bayes' rule: likelihood is not posterior probability . . . 7

3.4 Marginalization versus choosing the best pose . . . 7

4 Why 2D projections contain 3D information **7**

4.1 The projection operator . . . 7

4.2 A derivation of the central-slice theorem . . . 8

4.3 What does backprojection do? . . . 9

4.4 Why are many orientations necessary? . . . 9

4.5 Why 2D images of unknown orientations are harder than medical tomography . . . 9

5 The inverse problem: reconstructing a volume from known poses **10**

5.1 Linear algebra description . . . 10

5.2 Why regularization is unavoidable . . . 10

5.3 CTF-weighted reconstruction . . . 11

5.4 What if the CTF has a zero? . . . 11

6 Unknown poses: alignment and homogeneous refinement **11**

6.1 Pose estimation by projection matching . . . 11

6.2 Why reconstruction and alignment must be iterated . . . 11

6.3 Why the first map matters: the ab initio problem . . . 12

6.4 What does branch and bound actually accelerate? . . . 12

6.5 Gold-standard half-map refinement . . . 12

6.6 Homogeneous versus non-uniform refinement . . . 12

7 Mathematical prerequisite III: covariance, eigenvectors, and PCA 13
7.1 Why PCA appears in the study of molecular flexibility . . . 13
7.2 Mean and covariance of a multivariate dataset . . . 13
7.3 Deriving the first principal component . . . 14
7.4 Eigenvalues and explained variance . . . 14
7.5 What is the relationship between PCA and SVD? . . . 15
7.6 Probabilistic PCA . . . 15

8 Discrete classification: what if there are several structures? 16
8.1 What is a "class" in structural biology? . . . 16
8.2 A mixture model of K classes . . . 16
8.3 Soft class membership and Bayes' rule . . . 16
8.4 Why classification and alignment interact . . . 17

9 Expectation–maximization: derive the classification algorithm 17
9.1 Why do we need an iterative algorithm? . . . 17
9.2 The complete-data likelihood . . . 17
9.3 E-step: compute responsibilities . . . 18
9.4 M-step: update the classes . . . 18
9.5 Why EM increases likelihood, and what it does not guarantee . . . 18
9.6 How the cryo-EM E-step differs . . . 18
9.7 CryoSPARC: Heterogeneous Refinement versus 3D Classification . . . 19

10 Continuous heterogeneity: from PCA to 3DVA and RECOVAR 19
10.1 Why a finite number of classes may be the wrong model . . . 19
10.2 Deriving the continuous cryo-EM observation model . . . 20
10.3 What does 3DVA optimize? . . . 20
10.4 Why is a PCA coordinate not automatically a mechanical coordinate? . . . 20
10.5 Covariance is not a free-energy landscape . . . 22
10.6 What RECOVAR contributes . . . 22
10.7 The crucial distinction: PCA, classification, and covariance estimation . . . 22

11 Nonlinear models: cryoDRGN and 3DFlex 23
11.1 Why introduce a neural network at all? . . . 23
11.2 cryoDRGN: density conditioned on latent structure . . . 23
11.3 3DFlex: deform a canonical density rather than regenerate density . . . 23
11.4 Comparing assumptions fairly . . . 24
11.5 A subtle identifiability failure: pose versus deformation . . . 24

12 A practical CryoSPARC processing roadmap 24

12.1 From corrected movies to particle stacks . . . 24
12.2 First obtain a credible consensus structure . . . 24
12.3 Then ask what kind of heterogeneity is plausible . . . 24
12.4 Where 2D classification fits conceptually . . . 25
12.5 A reproducibility checklist . . . 25

13 A critical case study: flexible PP2A and DNA-spring particles 25
13.1 Why small flexible molecules are particularly hard . . . 25
13.2 A DNA spring creates a potential intervention—not an automatic force measurement . . . 25
13.3 Why two conditions can differ even without different conformations . . . 26
13.4 A validation ladder for force-dependent structural claims . . . 26
13.5 How should one select an analysis method? . . . 26

14 Worked statistical problems that connect the theory 26
14.1 How image averaging improves signal-to-noise . . . 26
14.2 Posterior uncertainty and confidence . . . 27
14.3 Population fractions and free energy . . . 27
14.4 How hard assignments underestimate ambiguity . . . 27
14.5 Why more PCA dimensions may help and hurt . . . 27

15 Self-test exercises with worked solutions 27

16 Summary: six statements worth retaining 29

Notation reference 30

Primary literature and software documentation 30

How to use this chapter

The preceding chapter established how the specimen potential modifies an electron wave, how an objective lens turns phase into measurable contrast, and why the recorded image is multiplied by a contrast transfer function (CTF) in Fourier space. This chapter begins at that recorded image. Its objective is to answer two linked questions: *How can thousands of noisy 2D projections reveal one 3D molecule?* and *How can a mixture of molecular conformations be separated without inventing structure from noise?*

No previous knowledge of probability, PCA, Bayesian inference, or the expectation–maximization algorithm is assumed. Sections 1–3 construct the mathematical toolkit. Sections 4–7 derive homogeneous reconstruction and refinement. Sections 8–11 derive discrete and continuous heterogeneity analysis. The last sections turn the equations into a CryoSPARC workflow and a critical validation checklist. Figures are original pedagogical illustrations rather than experimental datasets. Numerical values explicitly marked as illustrative are not instrument measurements.

Key concept

The unifying inverse problem. The microscope runs a *forward* process: choose a 3D structure, orient it, project it, apply the CTF, and add noise. Reconstruction attempts to invert that process, even though the structure and each particle's orientation are unknown. Classification adds another unknown: the molecule may not have had the same structure in every particle.

1 A bird's-eye view of the reconstruction problem

1.1 What data do we actually have?

After motion correction, CTF estimation, and particle extraction, an experiment typically provides many square image patches $I_1, \ldots, I_N$. Each patch is the noisy image of one molecular copy in a particular orientation. A molecule can appear entirely different when viewed from the side, front, or top. Cryo-EM avoids the need to physically rotate a single molecule through known angles: different molecules frozen in the ice naturally exhibit different orientations [1].

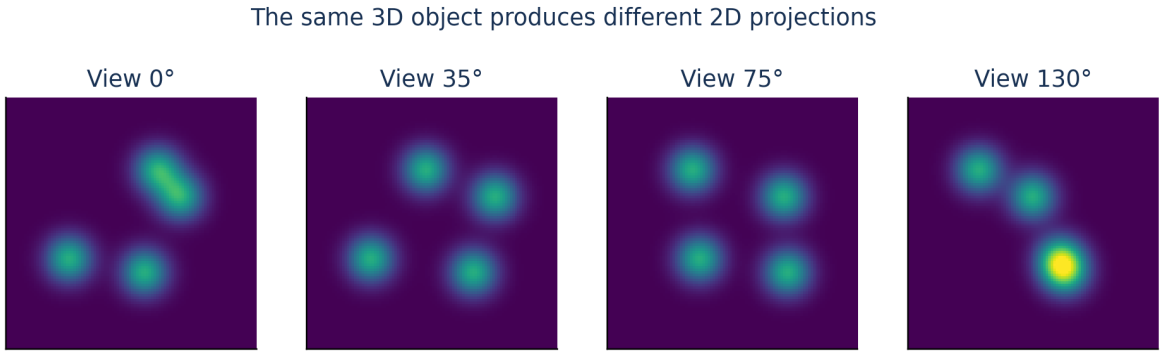


Figure 1: Four clean projections of one synthetic 3D point-cloud object. They differ because the object has been rotated, not because its composition changed. These are noise-free pedagogical examples; real particles are much less recognizable.

1.2 One formula for the entire imaging process

Write $V(\boldsymbol{r})$ for the three-dimensional Coulomb-potential-derived volume, $R_i \in SO(3)$ for the molecular orientation, $\boldsymbol{t}_i \in \mathbb{R}^2$ for its image-plane translation, and $C_i(\boldsymbol{q})$ for the CTF of particle i. We will use

$$Y_i(\boldsymbol{q}) = C_i(\boldsymbol{q})\, e^{-2\pi i \boldsymbol{q}\cdot\boldsymbol{t}_i}\, \widehat{V}\left(R_i^T \begin{bmatrix} q_x \\ q_y \\ 0 \end{bmatrix} \right) + \varepsilon_i(\boldsymbol{q}). \tag{1}$$

Y_i is the two-dimensional Fourier transform of the observed, mean-subtracted particle image, $\widehat{V}$ is the three-dimensional Fourier transform of V, and ε_i is noise. This is a simplified projection model: additional terms can be needed for optical aberrations, magnification, per-particle scaling, Ewald-sphere curvature, dose effects, or thickness-dependent multiple scattering.

Key concept

The unknowns have different roles. V is a *shared* variable because all particles contribute to the same reconstruction. The rotation R_i and translation $\boldsymbol{t}_i$ are *particle-specific nuisance variables*. In classification, each particle also receives a class label k_i or continuous conformation z_i.

1.3 What counts as successful reconstruction?

A map is not trustworthy merely because it looks protein-shaped. Success means that the recovered 3D volume predicts the measured images, has sufficient independent angular information, reproduces in separately processed half-datasets, and contains structural detail supported by the data rather than by the starting model. This motivates likelihood-based inference, regularization, and half-map validation [4, 7].

2 Mathematical prerequisite I: vectors, Fourier transforms, and rotations

2.1 Vectors, matrices, and inner products

A 2D image with $n \times n$ pixels can be flattened into a vector $\boldsymbol{y} \in \mathbb{R}^{n^2}$. A 3D volume with m^3 voxels becomes $\boldsymbol{v} \in \mathbb{R}^{m^3}$. A linear imaging operator A_i maps a volume to an image:

$$\boldsymbol{y}_i = A_i \boldsymbol{v} + \boldsymbol{\varepsilon}_i. \tag{2}$$

The operator A_i includes projection, translation, CTF filtering, interpolation, and sometimes masks or sampling. If A_i is represented as a matrix, it has n^2 rows and m^3 columns. In practice, this gigantic matrix is never explicitly stored. Fast projection and backprojection routines apply it implicitly.

The dot product $\boldsymbol{a}^T\boldsymbol{b} = \sum_j a_j b_j$ measures alignment between real vectors. The Euclidean norm is $\|\boldsymbol{a}\|^2 = \boldsymbol{a}^T\boldsymbol{a}$. The adjoint A_i^* reverses the direction of a linear mapping in the inner-product sense, and is central to reconstruction: its application resembles backprojecting an image into a volume.

2.2 What does a Fourier transform measure?

For $f(x, y)$, define

$$\widehat{f}(q_x, q_y) = \iint f(x, y) e^{-2\pi i (q_x x + q_y y)}\, \mathrm{d}x\, \mathrm{d}y. \tag{3}$$

The Fourier transform expresses the image as a sum of spatial waves. Low frequencies describe large shapes, high frequencies describe fine detail. A structural spacing d corresponds to spatial frequency $q \simeq 1/d$; for example, $d = 4$ Å corresponds to $q = 0.25$ Å^{-1}.

A spatial shift produces a Fourier-space phase:

$$\mathcal{F}\{f(\boldsymbol{r}-\boldsymbol{t})\}(\boldsymbol{q}) = e^{-2\pi i \boldsymbol{q}\cdot\boldsymbol{t}}\widehat{f}(\boldsymbol{q}). \tag{4}$$

This is why estimating an image translation can be implemented by varying a complex Fourier phase factor instead of repeatedly resampling the real-space image.

2.3 What is a 3D rotation?

A rotation is a 3×3 matrix R satisfying $R^T R = I$ and $\det R = 1$. The collection of such matrices is the group $SO(3)$. A particle orientation needs three angular degrees of freedom, often parameterized by Euler angles. An in-plane shift adds two more degrees of freedom. Thus the search is usually described as a five-dimensional *pose* problem [5].

If the molecular volume after rotation is $V_R(\boldsymbol{r}) = V(R^T\boldsymbol{r})$, the inverse rotation R^T appears because we ask: which original-molecule coordinate occupies the laboratory position $\boldsymbol{r}$? This convention will also explain the R^T in Eq. (1).

Worked example

Fourier shifts without changing amplitudes. Let a one-dimensional image contain a sinusoidal feature at $q = 0.20\,\text{pixels}^{-1}$, shifted by $t = 2$ pixels. Equation (4) multiplies its Fourier component by

$$e^{-2\pi i q t} = e^{-0.8\pi i}.$$

The complex phase changes by -0.8π radians, but the magnitude stays 1. In Fourier alignment, translation is a phase-estimation problem, not a change of the underlying molecular structure.

3 Mathematical prerequisite II: probability and inference

3.1 Random variables, distributions, and expectations

A random variable X is a quantity whose outcome is uncertain. A probability density $p(x)$ describes the relative frequency of nearby values. It obeys $p(x) \geq 0$ and $\int p(x)\,\mathrm{d}x = 1$. An expected value is

$$\mathbb{E}[X] = \int x\,p(x)\,\mathrm{d}x, \qquad \mathrm{Var}(X) = \mathbb{E}[(X-\mathbb{E}X)^2].$$

For two variables, $\mathrm{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}X)(Y-\mathbb{E}Y)]$. In cryo-EM the structural signal is unknown, the noise is random, and the pose and conformation can be treated as unknown latent variables.

3.2 Gaussian noise and why least squares appears

A scalar Gaussian distribution has density

$$p(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right]. \tag{5}$$

For d independent real-valued observations sharing noise variance σ^2, their joint log likelihood is

$$\log p(\boldsymbol{y} \mid \boldsymbol{\mu}) = -\frac{1}{2\sigma^2}\|\boldsymbol{y}-\boldsymbol{\mu}\|^2 - \frac{d}{2}\log(2\pi\sigma^2). \tag{6}$$

Maximizing this log likelihood is equivalent to minimizing squared error. Cryo-EM implementations often use frequency-dependent noise and complex-valued Fourier data; the same idea leads to an inverse-variance-weighted residual. The Gaussian approximation is a model for processed noise, not a claim that raw electron detection is Gaussian at arbitrarily low counts.

Worked example

Likelihood means "how well would this explanation generate the observation?" A measured pixel has $y = 1.2$ and noise standard deviation $\sigma = 0.5$. Candidate model A predicts $\mu_A = 1.0$, whereas model B predicts $\mu_B = 2.0$. Their likelihood ratio is

$$\frac{p(y \mid A)}{p(y \mid B)} = \exp\left[\frac{(1.2-2)^2 - (1.2-1)^2}{2(0.5)^2}\right] = e^{1.2} \approx 3.32.$$

The observed pixel is about 3.3 times more likely under A than B. Many pixels contribute evidence through sums of log likelihoods.

3.3 Bayes' rule: likelihood is not posterior probability

Bayes' rule is

$$\boxed{p(\theta \mid y) = \frac{p(y \mid \theta)p(\theta)}{p(y)}.} \tag{7}$$

Here $p(y \mid \theta)$ is the *likelihood*, $p(\theta)$ is the *prior*, and $p(\theta \mid y)$ is the *posterior*. A prior may express that very large high-frequency map amplitudes are unlikely; it can help stabilize reconstruction in weakly constrained frequency bands.

The maximum-likelihood estimate is $\widehat{\theta}_{\mathrm{ML}} = \arg\max_\theta p(y \mid \theta)$. The maximum-a-posteriori estimate is $\widehat{\theta}_{\mathrm{MAP}} = \arg\max_\theta p(y \mid \theta)p(\theta)$. MAP can be written as a regularized optimization problem because $-\log p(\theta)$ acts like a penalty [3, 4].

3.4 Marginalization versus choosing the best pose

Suppose the particle's orientation is unknown. A hard-alignment strategy finds a single best pose,

$$\widehat{R}_i = \arg\max_R p(y_i \mid R, V).$$

A Bayesian or likelihood-based marginalization instead averages over possible orientations:

$$p(y_i \mid V) = \int_{SO(3)} p(y_i \mid R, V)\, p(R)\, \mathrm{d}R, \tag{8}$$

with an additional integral or sum over shifts. This preserves uncertainty. It is especially valuable when particles are small and low-SNR, because two different poses may explain one image almost equally well [3, 16].

Key concept

Observed vs latent variables. The pixel values y_i are observed. Pose R_i, shift t_i, discrete class k_i, or continuous conformation z_i are latent: they must be inferred. The core statistical difficulty is that different latent-variable combinations may predict very similar noisy images.

4 Why 2D projections contain 3D information

4.1 The projection operator

Let $V(x, y, z)$ be a three-dimensional scalar function. Its projection along the z axis is

$$(PV)(x, y) = \int_{-\infty}^{\infty} V(x, y, z)\, \mathrm{d}z. \tag{9}$$

The integral adds contributions along an electron-beam path. In weak-phase imaging the projected specimen potential governs image formation; we then append the CTF and noise to create the actual measurement. To project from a different direction, rotate the volume first:

$$(P_R V)(x, y) = \int_{-\infty}^{\infty} V\Big(R^T (x, y, z)^T\Big)\, \mathrm{d}z.$$

Notice that PV is a function of two variables whereas V depends on three. A single projection has discarded depth information. Reconstruction depends on collecting many differently oriented projections.

4.2 A derivation of the central-slice theorem

The three-dimensional Fourier transform is

$$\widehat{V}(k_x, k_y, k_z) = \iiint V(x, y, z) e^{-2\pi i (k_x x + k_y y + k_z z)} \mathrm{d}x\mathrm{d}y\mathrm{d}z. \tag{10}$$

Now Fourier-transform the two-dimensional projection in Eq. (9):

$$\mathcal{F}_2\{PV\}(q_x, q_y) = \iint \left[\int V(x, y, z) \mathrm{d}z \right] e^{-2\pi i (q_x x + q_y y)} \mathrm{d}x\mathrm{d}y \tag{11}$$

$$= \iiint V(x, y, z) e^{-2\pi i (q_x x + q_y y + 0 z)} \mathrm{d}x\mathrm{d}y\mathrm{d}z \tag{12}$$

$$= \boxed{\widehat{V}(q_x, q_y, 0).} \tag{13}$$

Therefore the 2D Fourier transform of a projection is a plane through the origin of the 3D Fourier transform. For an arbitrary orientation,

$$\boxed{\mathcal{F}_2\{P_R V\}(\boldsymbol{q}) = \widehat{V}\Big(R^T (q_x, q_y, 0)^T\Big).} \tag{14}$$

This is the *Fourier or central-slice theorem*. It connects real-space line integration to sampling a plane in Fourier space. The proof assumes the appropriate integrability conditions and an ideal projection model; experimentally the CTF multiplies the slice and noise corrupts it. See the mathematical review by Singer and Sigworth [1].

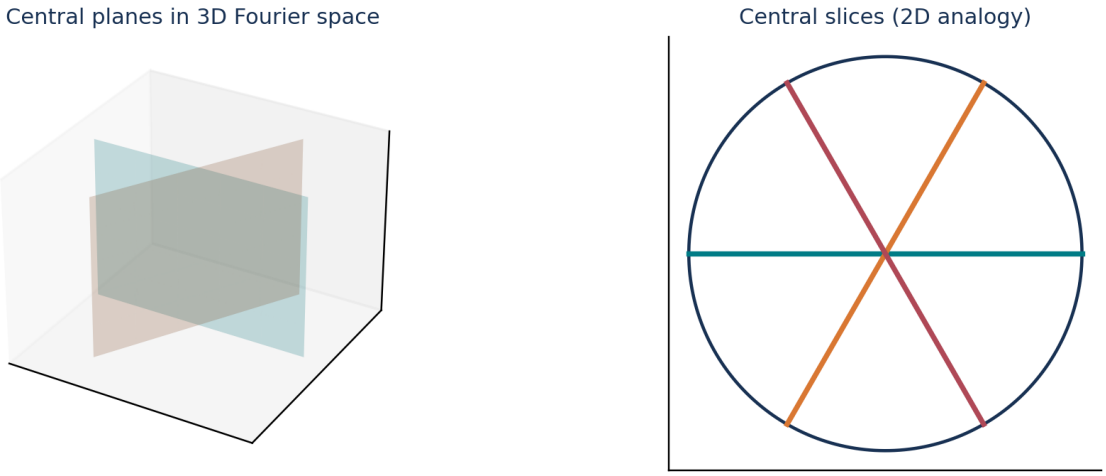


Figure 2: The Fourier-slice theorem. Rotating the specimen changes the orientation of the central plane sampled in the 3D Fourier transform. The right panel is a two-dimensional analogy showing angular coverage by differently oriented slices.

4.3 What does backprojection do?

Suppose the poses are known. In real-space backprojection, each image is spread backward along the ray directions from which it was observed. Summing these contributions produces a preliminary 3D volume. But ordinary unfiltered backprojection blurs low-frequency content because some Fourier samples are counted much more heavily than others.

In a 2D parallel-beam tomography analogue, filtered backprojection corrects this with a ramp filter:

$$f(x,y) = \int_0^{\pi} \left[\mathcal{R}_{1D}^{-1}\left\{|\omega|\widehat{p}_\theta(\omega)\right\}\right](x\cos\theta + y\sin\theta)\mathrm{d}\theta, \tag{15}$$

where p_θ is a measured projection at direction θ. This is an ideal continuous-angle formula. Single-particle 3D reconstruction instead typically uses Fourier gridding and/or optimization with CTF- and noise-dependent weighting, but the logic is similar: distribute each projection into the volume, then correct for uneven sampling.

4.4 Why are many orientations necessary?

If all particles appear in the same pose, all central slices lie in nearly the same plane. Even a million duplicates of that view cannot fully determine the missing 3D Fourier data. Increasing particle count improves noise averaging only in the frequencies and directions that are actually sampled.

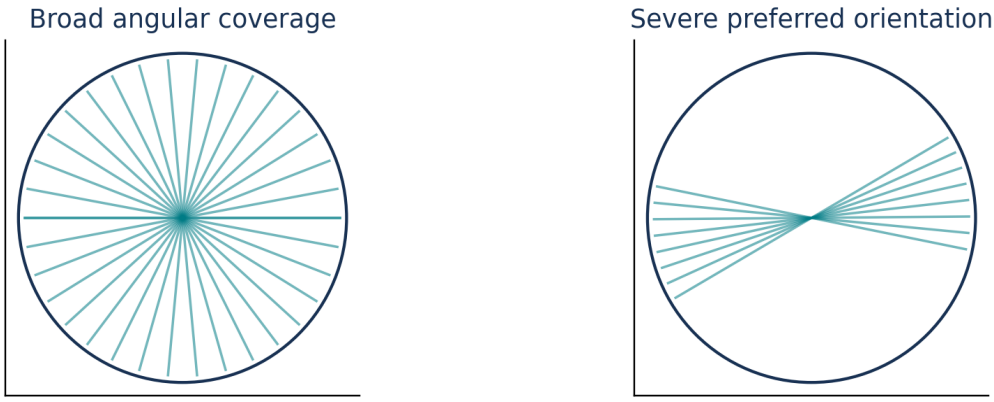


Figure 3: A two-dimensional frequency-space analogy. Broad angular coverage intersects many directions; preferred orientation repeatedly samples a narrow family of planes. The unsampled regions represent genuine missing information, not merely random noise.

Interpretation warning

Preferred orientation is not just "fewer particles." It creates direction-dependent information loss. A single global FSC resolution can look respectable while some directions have substantially poorer resolution. When studying a curved or flexible molecule, directional inadequacy may be mistaken for apparent bending or disappearance of density.

4.5 Why 2D images of unknown orientations are harder than medical tomography

In conventional computed tomography, acquisition angles are known from the scanning geometry. In single-particle cryo-EM, each molecular copy lands in an unknown orientation. Thus the inverse problem asks for both a 3D map and a set of orientations *at the same time*. Common-lines methods exploit intersections of Fourier slices; modern implementations frequently refine orientations iteratively using statistical comparison of experimental images to reference projections [2, 1].

Worked example

A numerical spatial-frequency limit. If the target feature size is $d = 3.5\,\text{Å}$, the corresponding frequency is $q = 1/3.5 \approx 0.286\,\text{Å}^{-1}$. If the image pixel size is $a = 1.0\,\text{Å}$, the Nyquist frequency is $q_N = 1/(2a) = 0.50\,\text{Å}^{-1}$. The feature is representable on the sampled image grid. If instead $a = 2.0\,\text{Å}$, then $q_N = 0.25\,\text{Å}^{-1}$ and this 3.5 Å feature is beyond Nyquist. No reconstruction algorithm can restore information excluded by the sampling limit without additional assumptions.

5 The inverse problem: reconstructing a volume from known poses

5.1 Linear algebra description

With poses held fixed, define A_i as the imaging operator (projection plus CTF plus shift). The measurement model is

$$\boldsymbol{y}_i = A_i \boldsymbol{v} + \boldsymbol{\varepsilon}_i.$$

Assume independent Gaussian pixel noise with covariance Σ_i. The negative log likelihood, up to constants, is

$$L(\boldsymbol{v}) = \frac{1}{2} \sum_{i=1}^{N} (\boldsymbol{y}_i - A_i \boldsymbol{v})^T \Sigma_i^{-1} (\boldsymbol{y}_i - A_i \boldsymbol{v}). \tag{16}$$

Taking a gradient with respect to $\boldsymbol{v}$ gives

$$\nabla L = \sum_i A_i^T \Sigma_i^{-1} (A_i \boldsymbol{v} - \boldsymbol{y}_i).$$

Setting this equal to zero yields the normal equations:

$$\boxed{\left(\sum_i A_i^T \Sigma_i^{-1} A_i \right) \widehat{\boldsymbol{v}} = \sum_i A_i^T \Sigma_i^{-1} \boldsymbol{y}_i.} \tag{17}$$

The right-hand side is a noise-weighted backprojection of the images. The left-hand matrix measures how strongly the data constrain different volume components. In practice these expressions are implemented using fast FFT, interpolation, and efficient GPU routines, not by forming the matrix explicitly.

5.2 Why regularization is unavoidable

If some spatial frequencies are weakly observed, Eq. (17) becomes unstable. Add a Gaussian prior

$$p(\boldsymbol{v}) \propto \exp\left[-\frac{\lambda}{2} \|L\boldsymbol{v}\|^2 \right].$$

The resulting MAP objective is

$$\widehat{\boldsymbol{v}} = \arg\min_{\boldsymbol{v}} \{ L(\boldsymbol{v}) + \tfrac{\lambda}{2} \|L\boldsymbol{v}\|^2 \},$$

with normal equations

$$\boxed{\left(\sum_i A_i^T \Sigma_i^{-1} A_i + \lambda L^T L \right) \widehat{\boldsymbol{v}} = \sum_i A_i^T \Sigma_i^{-1} \boldsymbol{y}_i.} \tag{18}$$

The regularizer L might favor smooth volumes or penalize implausible Fourier amplitudes. Bayesian approaches infer or adapt frequency-dependent priors rather than applying an arbitrary fixed filter [3, 4].

5.3 CTF-weighted reconstruction

If several measurements contribute to one ideal Fourier coefficient S, then $Y_i = C_i S + \varepsilon_i$. For independent Gaussian noise of variance σ_i^2 and a Gaussian prior $S \sim \mathcal{N}(0, \tau^2)$, the MAP estimator is

$$\boxed{\widehat{S} = \frac{\sum_i C_i^* Y_i / \sigma_i^2}{\sum_i |C_i|^2 / \sigma_i^2 + 1/\tau^2}.} \tag{19}$$

This is the principle behind CTF-aware reconstruction. It handles contrast reversals by using C_i^* and prevents an unstable division by an individual CTF zero.

Worked example

Combining two defocus images. Suppose the same Fourier coefficient S is observed with CTF factors $C_1 = 0.8$, $C_2 = -0.5$, and measured values $Y_1 = 0.88$, $Y_2 = -0.40$. Assume equal variances and neglect the prior. Then

$$\widehat{S} = \frac{0.8(0.88) + (-0.5)(-0.40)}{0.8^2 + (-0.5)^2} = \frac{0.904}{0.89} = 1.016.$$

The opposite signs do *not* cancel, because the reconstruction model explicitly includes the contrast reversals. Naive addition would have mixed them incorrectly.

5.4 What if the CTF has a zero?

If $C_i(q) = 0$ for an image, that observation contributes no first-order information to frequency q. Other images taken at different defocus may have nonzero transfer at the same frequency. If all observations have zero transfer in one direction/frequency, regularization can propose a smooth estimate, but it cannot manufacture experimental evidence. This distinction is essential when interpreting sharp-looking reconstructions.

6 Unknown poses: alignment and homogeneous refinement

6.1 Pose estimation by projection matching

Given a candidate volume V, generate synthetic projections from many orientations and shifts. Compare each predicted image $A(R,t)V$ to the actual particle y_i. Under white Gaussian noise, a maximum-likelihood pose approximately minimizes

$$\boxed{(\widehat{R}_i, \widehat{t}_i) = \arg\min_{R,t} \|y_i - A(R,t)V\|^2_{\Sigma_i^{-1}}.} \tag{20}$$

The notation $\|w\|^2_{\Sigma^{-1}} = w^T \Sigma^{-1} w$ is a weighted squared error. In Fourier space, the CTF and shift appear inside A, so the comparison is made against what the microscope would actually have observed, not an ideal blur-free projection.

6.2 Why reconstruction and alignment must be iterated

The volume is initially unknown, so we cannot accurately determine poses. But without poses, we cannot accurately reconstruct the volume. The solution is iterative refinement:

1. Start with a low-resolution volume $V^{(0)}$.
2. For every image, estimate possible poses under the current volume.
3. Reconstruct a new volume using those pose estimates and CTFs.

4. Repeat until held-out consistency and convergence criteria stabilize.

This alternating procedure is conceptually similar to coordinate descent or latent-variable optimization. Exact implementations differ: some maximize over poses, while others marginalize their uncertainty [16, 4].

6.3 Why the first map matters: the ab initio problem

A refinement algorithm can become trapped in a wrong solution if it begins from a poor initial model. Ab initio reconstruction attempts to obtain a viable initial volume without assuming a known structure. CryoSPARC's original approach combines coarse-to-fine branch-and-bound pose search with stochastic gradient descent (SGD) on volume parameters [5, 15]. In SGD, a mini-batch $\mathcal{B}$ approximates the full dataset gradient:

$$V^{(t+1)} = V^{(t)} - \eta_t \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \nabla_V \ell_i(V^{(t)}).$$

The random batch reduces computational cost and introduces stochasticity, although it does not guarantee finding the global optimum.

6.4 What does branch and bound actually accelerate?

A brute-force pose grid with N_R possible rotations and N_t shifts would require $N_R N_t$ image comparisons per particle. Hierarchical search first evaluates coarse orientation/shift regions, then refines only promising candidates. Bounds on the possible best score in a region let the algorithm discard regions that cannot win. Thus it avoids exhaustively evaluating every high-resolution pose [5, 15].

6.5 Gold-standard half-map refinement

Iterating a volume against noisy data creates a risk: noise can become reinforced by alignment to a reference that already contains the same noise. A standard defense is to split particles into two groups and independently refine two volumes V_A and V_B. Their reproducibility is tested using Fourier shell correlation (FSC):

$$\mathrm{FSC}(q) = \frac{\sum_{\mathbf{k} \in \mathcal{S}_q} \widehat{V}_A(\mathbf{k}) \widehat{V}_B(\mathbf{k})^*}{\sqrt{\sum_{\mathbf{k} \in \mathcal{S}_q} |\widehat{V}_A(\mathbf{k})|^2 \sum_{\mathbf{k} \in \mathcal{S}_q} |\widehat{V}_B(\mathbf{k})|^2}}. \tag{21}$$

Here $\mathcal{S}_q$ is a shell of Fourier frequencies near radius q; the reported FSC is ordinarily taken as the real shell correlation. The conventional 0.143 crossing gives a summary resolution $d \simeq 1/q$ under the gold-standard convention. It is not a universal threshold for every scientific question, and it cannot diagnose local resolution or directional bias by itself [7, 8].

Worked example

Interpreting an FSC crossing. Suppose the two half maps have FSC = 0.143 at $q = 0.250\ \text{Å}^{-1}$. The nominal global resolution is $d = 1/q = 4.0$ Å. This does not mean every residue is resolved at 4.0 Å: a flexible domain may have local resolution of 8 Å, and a preferred-orientation dataset may have worse resolution in certain directions.

6.6 Homogeneous versus non-uniform refinement

Homogeneous refinement assumes a single consensus structure. Non-uniform refinement incorporates spatially adaptive regularization to prevent flexible or disordered regions from dominating the result and to preserve useful signal in better-ordered regions. It is an improved way to estimate a consensus map; it does *not* by itself establish a unique conformational distribution [6].

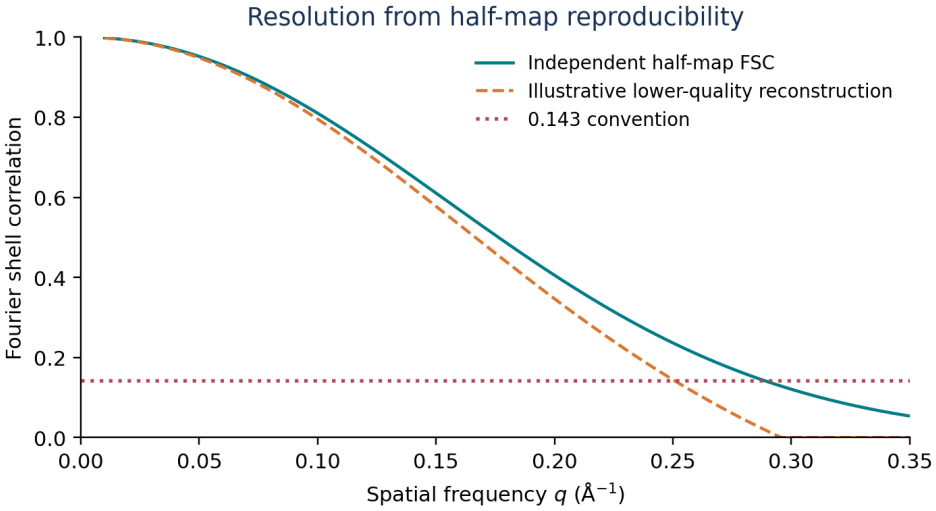


Figure 4: Illustrative FSC curves. Each point compares Fourier coefficients of independently refined half maps within a shell. The 0.143 crossing is one conventional global-resolution criterion, not a statement that all regions and directions are resolved equally.

Interpretation warning

Reference bias and circular reasoning. A map that agrees with the starting reference is not automatically independent evidence for the reference. Evaluate sensitivity to initialization, use appropriately low-pass-filtered starting models, maintain half-set independence, and check that new structural features are supported by experimental images.

7 Mathematical prerequisite III: covariance, eigenvectors, and PCA

7.1 Why PCA appears in the study of molecular flexibility

Imagine that every particle belonged to a slightly different conformation. Even when we can reconstruct an average map, it may conceal those differences. We want a small number of coordinates summarizing the main directions in which structure changes. Principal component analysis (PCA) is the simplest systematic approach. But before explaining 3DVA or RECOVAR, we should derive PCA from elementary statistics [10].

7.2 Mean and covariance of a multivariate dataset

Suppose each observation is a vector $x_i \in \mathbb{R}^D$. Its sample mean is

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i.$$

Its sample covariance matrix is

$$\boxed{\Sigma = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})(x_i - \bar{x})^T.} \tag{22}$$

The diagonal entry Σ_{jj} measures the variance of feature j. Off-diagonal Σ_{jk} measures how two features tend to move together. A positive covariance means that when one coordinate is above its mean, the other tends to be above its mean too.

7.3 Deriving the first principal component

Take a unit direction $u \in \mathbb{R}^D$, with $u^T u = 1$. The scalar coordinate of observation x_i along that direction is

$$z_i = u^T (x_i - \bar{x}).$$

Its sample variance is

$$\mathrm{Var}(z) = u^T \Sigma u.$$

PCA chooses the direction maximizing this variance:

$$\max_u u^T \Sigma u \quad \text{subject to} \quad u^T u = 1. \tag{23}$$

Introduce a Lagrange multiplier λ:

$$\mathcal{L}(u, \lambda) = u^T \Sigma u - \lambda (u^T u - 1).$$

Setting the derivative to zero,

$$2\Sigma u - 2\lambda u = 0,$$

so

$$\boxed{\Sigma u = \lambda u.} \tag{24}$$

The maximizing direction is the eigenvector with the largest eigenvalue. The second component is the orthogonal eigenvector with the next largest eigenvalue, and so on.

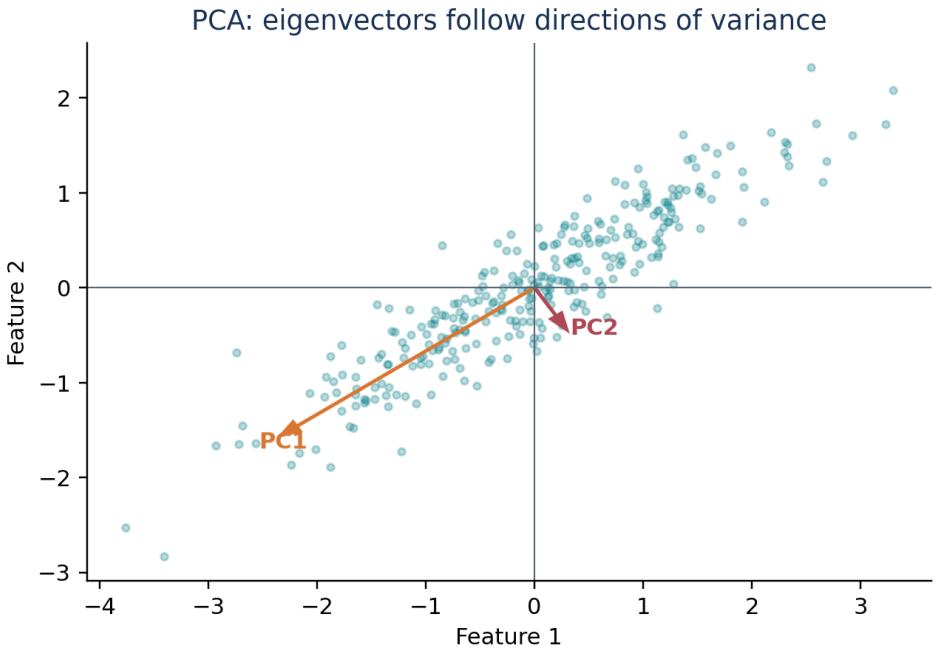


Figure 5: A two-dimensional dataset and its covariance eigenvectors. PC1 is the direction of largest variation, whereas PC2 is orthogonal and captures less variation. PCA finds directions of spread; it does not, by itself, find biological states or clusters.

7.4 Eigenvalues and explained variance

If $\lambda_1 \geq \cdots \geq \lambda_D \geq 0$ are covariance eigenvalues, then the proportion explained by the first K principal components is

$$\boxed{\mathrm{EVR}(K) = \frac{\sum_{j=1}^{K} \lambda_j}{\sum_{j=1}^{D} \lambda_j}.} \tag{25}$$

A PCA reconstruction using K components has the form

$$x_i \approx \bar{x} + \sum_{j=1}^{K} z_{ij} u_j .$$

In a molecular interpretation, $\bar{x}$ could represent the average 3D map, u_j a 3D difference map, and z_{ij} a particle-specific variability coordinate. Note that when x_i is an image, it is generally *not* correct to perform ordinary PCA directly on raw 2D pixels and interpret the resulting components as 3D molecular motion: changing orientations, CTFs, and noise all generate enormous image variance.

Worked example

Compute PCA entirely by hand. Consider four observations

$$(2,0), \quad (-2,0), \quad (0,1), \quad (0,-1).$$

Their mean is $(0,0)$. The sample covariance is

$$\Sigma = \frac{1}{3}\begin{bmatrix} 8 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 8/3 & 0 \\ 0 & 2/3 \end{bmatrix}.$$

Its eigenvectors are $u_1 = (1,0)^T$ and $u_2 = (0,1)^T$, with eigenvalues $\lambda_1 = 8/3$ and $\lambda_2 = 2/3$. The variance explained by PC1 is

$$\frac{8/3}{8/3 + 2/3} = \frac{8}{10} = 80\%.$$

The first point $(2,0)$ has PCA coordinates $(z_1, z_2) = (2,0)$. Compressing to one dimension retains 80% of the dataset's total sample variance, even though that does not make the coordinate a biological reaction coordinate.

7.5 What is the relationship between PCA and SVD?

Form the centered data matrix $X \in \mathbb{R}^{N \times D}$ whose rows are $x_i - \bar{x}$. Singular value decomposition (SVD) gives

$$X = U \operatorname{diag}(s_1, \ldots, s_r)\, W^T.$$

Because $\Sigma = X^T X/(N-1)$, its eigenvectors are the columns of W, and the covariance eigenvalues are $\lambda_j = s_j^2/(N-1)$. SVD is often the computational route to PCA for moderate-sized datasets. In cryo-EM, storing an explicit covariance over millions of voxels is impossible, so methods exploit imaging structure, low-rank approximations, and regularization.

7.6 Probabilistic PCA

Ordinary PCA is deterministic. Probabilistic PCA introduces latent Gaussian coordinates:

$$x_i = \mu + W z_i + \epsilon_i, \qquad z_i \sim \mathcal{N}(0, I), \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2 I). \tag{26}$$

The observation covariance is

$$\operatorname{Cov}(x) = WW^T + \sigma^2 I.$$

Thus PCA-like subspaces can arise from fitting a latent-variable likelihood. This link matters because cryo-EM does not observe volumes x_i directly: it observes noisy, CTF-filtered, differently oriented projections of them. Probabilistic reasoning makes it possible to infer covariance while accounting for those projections [10, 12].

8 Discrete classification: what if there are several structures?

8.1 What is a "class" in structural biology?

A class is a set of particles modeled as sharing one three-dimensional state, up to rotation, translation, and imaging noise. The difference may be compositional (a domain or subunit is present/absent), conformational (open/closed), or an artifact (damaged particles or contaminants). A class label is *not* proof that molecules occupy a unique thermodynamic state.

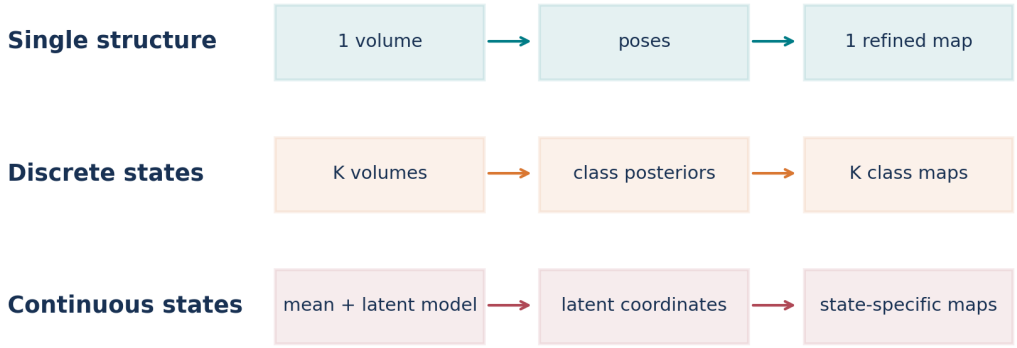


Figure 6: Three increasingly expressive modeling choices. Homogeneous reconstruction produces one map; discrete classification produces several class maps; continuous methods associate images with a low-dimensional distribution of volumes or deformations.

8.2 A mixture model of K classes

Let $V_1, \dots, V_K$ be class volumes and π_k their population fractions, with $\pi_k \geq 0$ and $\sum_k \pi_k = 1$. Each particle has latent class $k_i \in \{1, \dots, K\}$ and pose (R_i, t_i). Its marginal probability is

$$p(y_i \mid V_1, \dots, V_K) = \sum_{k=1}^{K} \pi_k \iint p(y_i \mid V_k, R, t)\, p(R, t \mid k)\, \mathrm{d}R\mathrm{d}t. \tag{27}$$

In words: each class proposes a structure, each pose projects it, and we average over the possibilities according to their probabilities. The complete dataset likelihood is $\prod_i p(y_i \mid \{V_k\})$ under the common conditional-independence model.

8.3 Soft class membership and Bayes' rule

If poses are known or temporarily fixed, the posterior probability of class k for particle i is

$$\boxed{\gamma_{ik} \equiv p(k_i = k \mid y_i) = \frac{\pi_k p(y_i \mid k)}{\sum_{\ell=1}^{K} \pi_\ell p(y_i \mid \ell)}.} \tag{28}$$

The values γ_{ik} are called *responsibilities*. They are between 0 and 1, and for each particle they sum to 1. A hard assignment picks $\arg\max_k \gamma_{ik}$; a soft assignment lets one ambiguous particle contribute to more than one class in proportion to its posterior probability.

Worked example

A numerical two-class Bayesian assignment. Imagine two one-dimensional Gaussian classes with equal prior probability, means $\mu_A = 0$ and $\mu_B = 2$, and shared standard deviation $\sigma = 1$. For an observed feature $x = 0.5$,

$$\frac{p(x \mid A)}{p(x \mid B)} = \exp\left[\frac{(0.5-2)^2 - (0.5-0)^2}{2}\right] = e^1.$$

With equal priors,

$$p(A \mid x) = \frac{e}{1+e} \approx 0.731, \quad p(B \mid x) \approx 0.269.$$

For $x = 1$, both probabilities are 0.5. Classification is therefore not always certain, even when the statistical model is correct.

8.4 Why classification and alignment interact

If a particle is rotated differently, its image changes. If the protein also changes conformation, its image changes. These two effects can be confounded. A wrong pose can be mistaken for a different class; conversely, a structural change can be incorrectly absorbed by a pose adjustment. Thus classification and orientation estimation should be considered jointly unless the poses are sufficiently reliable [3, 16].

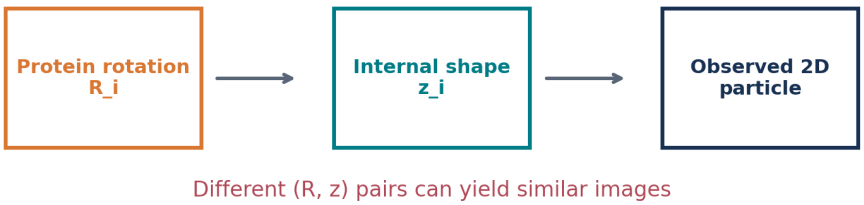


Figure 7: Rotation and internal conformation are separate latent variables. A single noisy 2D projection can make them difficult to distinguish, particularly for small, elongated, or nearly symmetric particles.

9 Expectation–maximization: derive the classification algorithm

9.1 Why do we need an iterative algorithm?

Mixture-model likelihoods contain sums over unknown classes and poses. Directly maximizing them over all volumes, labels, and rotations is difficult. The expectation–maximization (EM) algorithm alternates between estimating the latent-variable distribution and updating the model parameters. Dempster, Laird, and Rubin established its general mathematical framework [9].

9.2 The complete-data likelihood

For a simple Gaussian mixture with class means μ_k, priors π_k, and a known common variance σ^2, introduce binary indicators $z_{ik} \in \{0, 1\}$ with $\sum_k z_{ik} = 1$. The log likelihood as if all labels were known is

$$\log p(x, z \mid \theta) = \sum_{i,k} z_{ik} \left[\log \pi_k - \frac{(x_i - \mu_k)^2}{2\sigma^2} - \log(\sqrt{2\pi}\sigma)\right]. \tag{29}$$

We do not observe z_{ik}, so EM replaces them by their conditional expectations.

9.3 E-step: compute responsibilities

Given current parameters $\theta^{(t)}$, compute

$$\boxed{\gamma_{ik}^{(t)} = \frac{\pi_k^{(t)} \exp[-(x_i - \mu_k^{(t)})^2/(2\sigma^2)]}{\sum_\ell \pi_\ell^{(t)} \exp[-(x_i - \mu_\ell^{(t)})^2/(2\sigma^2)]}.} \tag{30}$$

The E-step does not physically relabel particles. It calculates the probability of each explanation under the current model.

9.4 M-step: update the classes

Holding the responsibilities fixed, maximize the expected complete log likelihood. For the Gaussian mixture,

$$\boxed{\pi_k^{(t+1)} = \frac{1}{N}\sum_i \gamma_{ik}^{(t)}, \qquad \mu_k^{(t+1)} = \frac{\sum_i \gamma_{ik}^{(t)} x_i}{\sum_i \gamma_{ik}^{(t)}}.} \tag{31}$$

The new class mean is a posterior-weighted average of observations; the class fraction is the average posterior membership. In cryo-EM, μ_k is replaced by a three-dimensional volume V_k, and the update is CTF-aware, pose-aware, and regularized.

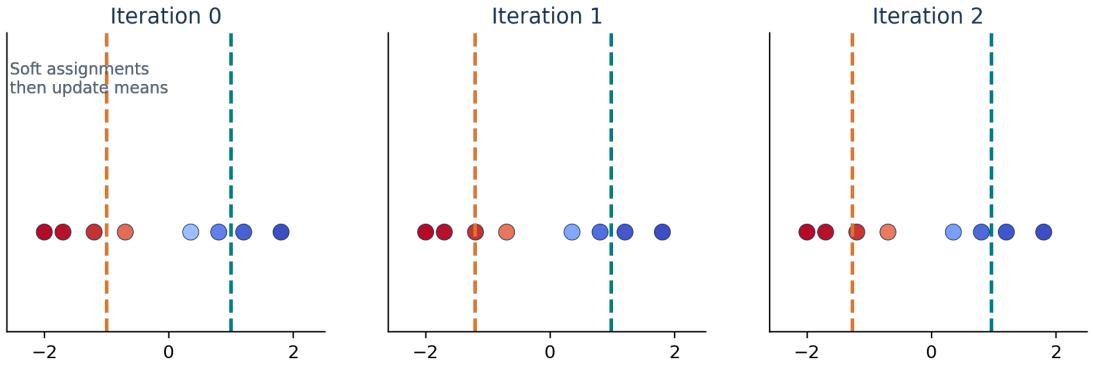


Figure 8: A toy expectation–maximization process. Each point is softly assigned to two Gaussian classes (color), and the dashed mean locations are updated using responsibility-weighted averages. Real cryo-EM EM operates over volumes, rotations, shifts, and CTFs.

9.5 Why EM increases likelihood, and what it does not guarantee

Define

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{z \mid x, \theta^{(t)}}[\log p(x, z \mid \theta)].$$

EM maximizes Q in the M-step. Jensen's inequality provides a lower-bound interpretation of the observed-data log likelihood, showing that exact EM steps do not decrease the likelihood under standard conditions. But a nondecreasing likelihood does not ensure the global optimum, the correct number of classes, or a physically meaningful decomposition. Initial conditions, regularization, and model mismatch still matter [9].

9.6 How the cryo-EM E-step differs

A cryo-EM latent state is typically (k, R, t) rather than only k. Its joint responsibility has the structure

$$w_i(k, R, t) \propto \pi_k \, p(R, t \mid k) \, \exp\left[-\frac{1}{2}\|y_i - A(R, t)V_k\|^2_{\Sigma_i^{-1}}\right]. \tag{32}$$

The E-step computes or approximates weights over many classes and poses. The M-step uses those weights to reconstruct new volumes. Hard pose maximization, pose marginalization, and hard versus soft class assignments are distinct approximations, each affecting speed and uncertainty [16, 4].

Worked example

One complete EM iteration. Consider observations $x = (-1, 0, 2)$, two classes, initial means $\mu_A = 0$, $\mu_B = 2$, shared $\sigma = 1$, and equal priors. The class-A responsibilities are

$$\gamma_A(-1) = \frac{1}{1+e^{-4}} = 0.982, \quad \gamma_A(0) = \frac{1}{1+e^{-2}} = 0.881, \quad \gamma_A(2) = \frac{1}{1+e^{2}} = 0.119.$$

Thus $N_A \approx 1.982$, and the M-step gives

$$\mu_A' = \frac{0.982(-1) + 0.881(0) + 0.119(2)}{1.982} \approx -0.375.$$

The new class fraction is $\pi_A' = 1.982/3 \approx 0.661$. Class B receives the complementary responsibilities. This is the exact numerical logic behind soft classification, although a real class update reconstructs a 3D map from many differently oriented particles.

9.7 CryoSPARC: Heterogeneous Refinement versus 3D Classification

The two jobs use different assumptions [16, 17, 18]. **Heterogeneous Refinement** jointly refines class volumes and particle orientations, using several initial 3D references. **3D Classification** uses input particle poses without realigning orientations or shifts while estimating class memberships and volumes. The latter is faster and effective when input poses are reliable; the former may correct pose errors but faces a harder joint optimization problem. These are not interchangeable simply because both output multiple maps.

Interpretation warning

Classification is not a proof of discrete thermodynamic basins. A continuum of conformations can be artificially partitioned into K classes. Conversely, two real states may merge into one average if the image information is weak. Class count and occupancy should be tested for robustness to masks, initialization, particle subsets, and model assumptions.

10 Continuous heterogeneity: from PCA to 3DVA and RECOVAR

10.1 Why a finite number of classes may be the wrong model

A protein with a flexible hinge may move through many conformations without occupying only two rigid states. A discrete mixture model can divide the movement into bins, but its output depends on the chosen number of classes. A continuous model instead introduces a latent coordinate $z_i \in \mathbb{R}^K$. Every particle may have its own structural realization. A simple linear model is

$$\boxed{V_i(\boldsymbol{r}) = V_0(\boldsymbol{r}) + \sum_{j=1}^{K} z_{ij} U_j(\boldsymbol{r}).} \tag{33}$$

V_0 is a mean or consensus volume; U_j are difference-volume directions; z_{ij} are the coordinates of particle i. This is the high-dimensional analogue of PCA. The difference directions can be interpreted as volumetric changes but are not necessarily literal atomic displacements.

10.2 Deriving the continuous cryo-EM observation model

Substitute Eq. (33) into the forward model, with poses treated as known for the moment:

$$y_i = A_i\left(V_0 + \sum_{j=1}^{K} z_{ij} U_j\right) + \varepsilon_i \tag{34}$$

$$= A_i V_0 + \sum_{j=1}^{K} z_{ij} A_i U_j + \varepsilon_i. \tag{35}$$

Let $U = [U_1 \ \cdots \ U_K]$. Then

$$\boxed{y_i = A_i V_0 + A_i U z_i + \varepsilon_i.} \tag{36}$$

This equation is central to 3D variability analysis (3DVA). The algorithm works from *images of projections*, not direct observations of the 3D conformational volumes. It estimates an informative low-dimensional volume subspace and coordinates while accounting for pose and CTF [11].

10.3 What does 3DVA optimize?

The method fits a linear subspace model of high-resolution variability to aligned particles. A schematic data-fit objective is

$$\min_{U,\{z_i\}} \sum_i \|y_i - A_i V_0 - A_i U z_i\|^2_{\Sigma_i^{-1}} + \mathcal{R}(U, z), \tag{37}$$

where $\mathcal{R}$ represents practical constraints and regularization. This equation conveys the model family, not the complete production implementation. In CryoSPARC, the job solves for orthogonal principal variability modes and exports particle coordinates and volume-mode information; its display job visualizes series of maps along chosen components [20, 11].

Each component U_j can contain positive and negative densities. A positive region indicates increasing density at that location as the coordinate grows; a negative region indicates decreasing density. That might correspond to physical motion, changes in occupancy, flexibility, or even residual alignment errors. It should not automatically be interpreted as atoms moving independently into regions of negative density.

10.4 Why is a PCA coordinate not automatically a mechanical coordinate?

Suppose a protein has a curved hinge trajectory, described in two structural coordinates by

$$x(\theta) = \cos\theta, \qquad y(\theta) = \sin\theta.$$

The true structural states lie on an arc of a circle, not a straight line. PCA finds a single straight axis through this arc. Projecting each point onto that line loses curvature information. For a sufficiently large arc, different conformations may receive similar projected coordinates.

Worked example

Two distinct structures with one PCA coordinate. Consider three points on a symmetric semicircle: $x_1 = (-1, 0)$, $x_2 = (0, 1)$, $x_3 = (1, 0)$. Their mean is $(0, 1/3)$ and the sample covariance is diagonal with $\lambda_x = 1$ and $\lambda_y = 1/3$. PC1 is horizontal. If we retain only PC1, the middle point has coordinate zero, while the vertical deviation of that state is lost. With other geometries, two distinct configurations can even have identical projections onto the leading axis. Thus a smooth movie along a principal mode does not uniquely identify a physical path or force coordinate.

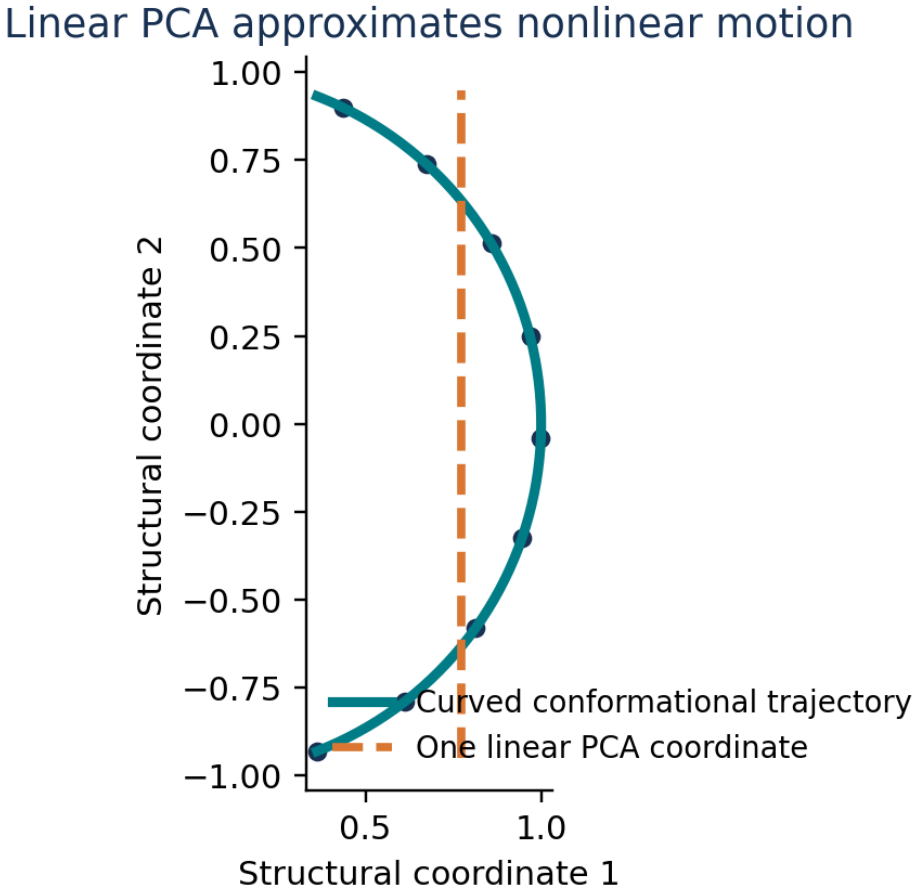


Figure 9: A nonlinear conformational trajectory (solid curve) and one linear PCA axis (dashed). Linear PCA may faithfully approximate small-amplitude motion but can distort or even fold a large nonlinear trajectory when used as a physical reaction coordinate.

10.5 Covariance is not a free-energy landscape

For a state variable q at thermal equilibrium, the density $p(q)$ is related to a free energy along that specified coordinate by

$$F(q) = -k_B T \log p(q) + \text{constant}. \tag{38}$$

But three conditions matter. First, the data must represent a stationary equilibrium distribution rather than a biased preparation or selection process. Second, q and its measure must be physically specified. Third, density estimation and particle-assignment uncertainty must be handled appropriately.

Under a nonlinear transformation $z' = g(z)$, probability densities obey

$$p_{z'}(z') = p_z(z) \left| \det \frac{\partial z}{\partial z'} \right|. \tag{39}$$

Thus $-k_B T \log p(z)$ shifts by a Jacobian term under coordinate changes. A general learned latent-space density cannot automatically be interpreted as a coordinate-invariant molecular potential of mean force. CryoSPARC's 3DFlex documentation highlights this problem, and RECOVAR explicitly seeks more interpretable density estimates in a PCA-based embedding [21, 12].

10.6 What RECOVAR contributes

RECOVAR (regularized covariance estimation and kernel regression) is a method for recovering continuous heterogeneity from noisy images [12]. It is grounded in a 3D covariance estimation problem rather than training a generic nonlinear neural decoder. The central statistical object is

$$\Sigma_V = \mathbb{E}[(V_i - \bar{V})(V_i - \bar{V})^T]. \tag{40}$$

Because V_i is not observed directly, recovering Σ_V requires correcting for the imaging operators A_i, noise, and limited Fourier sampling. The estimated covariance is regularized, its dominant eigenvectors define principal components, and images are assigned uncertain coordinates in that subspace.

The method also uses adaptive kernel regression to reconstruct maps corresponding to particular latent regions. Schematically, for a target coordinate z_*,

$$\widehat{V}(z_*) = \arg\min_V \sum_i w_i(z_*) \|y_i - A_i V\|^2_{\Sigma_i^{-1}} + \mathcal{R}(V), \tag{41}$$

where $w_i(z_*)$ assigns more weight to particles whose inferred conformations are near z_*. This is a conceptual form of the weighted reconstruction, not RECOVAR's complete implemented objective. Its paper studies conformational density, uncertainty, and low-free-energy paths, while noting the importance of embedding geometry [12].

10.7 The crucial distinction: PCA, classification, and covariance estimation

Method	Primary mathematical object	Typical output / interpretation
PCA	Eigenvectors of a covariance matrix	Orthogonal directions of variance, not class labels.
Discrete mixture	Several class likelihoods and priors	Soft/hard memberships in K modeled states.
3DVA	Linear subspace fit to CTF-filtered projections	Mean map, volume modes, particle variability coordinates.
RECOVAR	Regularized 3D covariance with coordinate inference	PCA embedding, uncertainty and locally reconstructed states.

11 Nonlinear models: cryoDRGN and 3DFlex

11.1 Why introduce a neural network at all?

If a protein undergoes bending, twisting, domain rotation, and flexible occupancy changes simultaneously, a small linear basis might not describe the volume family accurately. A nonlinear model instead writes

$$\boxed{V_i = G_\theta(z_i),} \tag{42}$$

where G_θ is a learned mapping from low-dimensional latent coordinates to a 3D density. Its observation model is

$$y_i = A_i G_\theta(z_i) + \varepsilon_i.$$

The challenge is that infinitely many choices of G_θ and latent coordinates can fit noisy observations. Strong inductive biases, architecture constraints, regularization, and independent validation are required.

11.2 cryoDRGN: density conditioned on latent structure

cryoDRGN represents structural heterogeneity using a neural generative model of volume density conditioned on a latent coordinate. A variational encoder infers a distribution $q_\phi(z_i \mid y_i)$, while a decoder describes 3D density as a function of 3D position and z_i [13]. Schematically,

$$\rho_\theta(\boldsymbol{r}, z_i) = \mathrm{Decoder}_\theta(\boldsymbol{r}, z_i). \tag{43}$$

A typical variational objective has the evidence-lower-bound structure

$$\mathcal{L}_{\mathrm{ELBO}} = \mathbb{E}_{q_\phi(z \mid y)}[\log p_\theta(y \mid z)] - D_{\mathrm{KL}}\big(q_\phi(z \mid y) \| p(z)\big)\,. \tag{44}$$

Here D_{KL} is Kullback–Leibler divergence. For discrete variables,

$$D_{\mathrm{KL}}(q\|p) = \sum_z q(z) \log \frac{q(z)}{p(z)},$$

with an analogous integral for continuous variables. The first term rewards reproducing images; the second regulates the inferred latent distribution.

Key concept

What a neural latent coordinate means. A coordinate z_i is a location in a learned representation. Its numerical scale, axes, and density may be arbitrary under transformations that can be compensated by the decoder. It should not be reported as a bending angle, force, or free energy unless an independent, calibrated mapping justifies that interpretation.

11.3 3DFlex: deform a canonical density rather than regenerate density

3DFlex uses a different physical prior: the molecule's conformations are related by a smooth deformation of one canonical 3D map V_0. A latent coordinate generates a displacement field

$$u_\theta(\boldsymbol{r}, z_i),$$

which is used to transport/convect the canonical density into the state V_i [14]. In a simple illustrative pullback notation,

$$V_i(\boldsymbol{r}) \approx V_0(\boldsymbol{r} - u_\theta(\boldsymbol{r}, z_i)), \tag{45}$$

although real implementations use a specific flow/convection operator and deformation mesh. The network learns a deformation field, not an unconstrained density for every state. A tetrahedral mesh and rigidity prior discourage physically implausible local warping [21].

This is well suited to coherent motion of domains. It is less naturally suited to compositional changes: density that genuinely appears or disappears cannot be represented cleanly by simply moving existing density around. 3DFlex can also generate interpolated shapes between observed endpoints even when no real particles support the intermediate conformations. This is a model-implied interpolation, not experimental evidence of occupancy [21].

11.4 Comparing assumptions fairly

Approach	Assumed representation	Principal caution
3D classification	K discrete 3D volumes	A continuum may be split arbitrarily.
3DVA	$V(z) = V_0 + Uz$	Large nonlinear motions may be distorted.
RECOVAR	Covariance/PCA embedding; local reconstruction	Pose errors and embedding/selection bias remain relevant.
cryoDRGN	Nonlinear learned density $G_\theta(z)$	Latent geometry and density need not be physical.
3DFlex	Warped canonical density via flow field	Cannot cleanly add/remove density; interpolation can be unsupported.

11.5 A subtle identifiability failure: pose versus deformation

Suppose the protein is an elongated object. Rotating it slightly can shift the apparent position of its ends within a 2D image. Bending the protein can also shift those ends. A model might explain an observed difference through either route. If we change the inferred pose and the conformation simultaneously, two parameter settings can produce nearly identical images:

$$A(R_1)V(z_1) \approx A(R_2)V(z_2).$$

This is a failure of identifiability. More training data, alternative views, appropriate structural constraints, and held-out tests may help, but a flexible model alone cannot guarantee a unique solution.

12 A practical CryoSPARC processing roadmap

12.1 From corrected movies to particle stacks

A typical workflow proceeds through (i) motion correction to reduce beam-induced movement, (ii) CTF estimation per micrograph, (iii) particle picking and extraction, and (iv) 2D classification to discard contamination or obviously poor particle images. The output is a curated stack of small 2D images with associated pixel sizes, micrograph identities, and CTF parameters. Different software versions may rename or modify these jobs; the linked documentation is the source of truth for exact settings [15, 16].

12.2 First obtain a credible consensus structure

Run ab initio reconstruction to obtain a low-resolution initial map. For a homogeneous dataset, follow with Homogeneous Refinement or Non-Uniform Refinement, preserving gold-standard half-sets. Inspect particle orientation distributions, the real-space map, FSC, local resolution, and any model-induced artifacts. A single high-looking nominal resolution number is insufficient [5, 6, 19].

12.3 Then ask what kind of heterogeneity is plausible

If the sample contains extra subunits, degraded particles, or alternative complexes, begin with discrete classification or heterogeneous refinement. If the sample contains coherent smooth motions, use 3DVA to

inspect variability and possibly RECOVAR or other continuous approaches. If the motion appears large and nonlinear, a deformation model such as 3DFlex or a flexible density model such as cryoDRGN may be appropriate. These are choices about the biological hypothesis and forward-model assumptions, not a ladder of universally better algorithms.

12.4 Where 2D classification fits conceptually

CryoSPARC's 2D Classification finds recurring, aligned image patterns by learning class averages and pose/shift memberships. Unlike 3D classification, the underlying class templates are two-dimensional. A side view and a top view of the *same* molecular structure can therefore occupy different 2D classes. One must never interpret distinct 2D classes as necessarily distinct 3D conformations.

12.5 A reproducibility checklist

For every reported state, document the initial particle count, excluded-particle decisions, CTF/defocus quality, orientation distribution, mask design, classification settings, reference initialization, half-map FSC, relevant map-to-model agreement, and whether the class/state reappears under reasonable alternative settings. For mechanistic claims, consider the stronger tests in the next section.

13 A critical case study: flexible PP2A and DNA-spring particles

13.1 Why small flexible molecules are particularly hard

The isolated PP2A A-subunit scaffold is roughly 65 kDa and consists of fifteen HEAT repeats. Its elongated, curved architecture can change shape through inter-repeat motions. For single-particle cryo-EM, this combines several challenges: modest scattering signal per particle, non-spherical geometry, related repeat motifs, substantial internal flexibility, and possible density from a tethered DNA spring. A weakly resolved 2D image may not uniquely distinguish the object's orientation from its conformation.

Consider a particle with two possible bending coordinates, z_1 and z_2. If the protein is oriented nearly end-on, bending toward or away from the electron beam can appear as almost no change in the projected image. The model may assign an uncertain z_1 even when the molecule is genuinely bent. In another view the same deformation may be large and visible. Thus the information carried by each image about a conformation depends on orientation and CTF, not just on the intrinsic amplitude of motion.

13.2 A DNA spring creates a potential intervention—not an automatic force measurement

Suppose experimental conditions c distinguish relaxed ssDNA from a hybridized duplex spring. A useful generative model is

$$p(y_i \mid c_i) = \int p(y_i \mid z, R, t, c_i)\, p(z, R, t \mid c_i)\, \mathrm{d}z\mathrm{d}R\mathrm{d}t. \tag{46}$$

The condition label c_i is known from specimen preparation; the conformation and pose are unknown. The biological question is whether $p(z \mid c)$ changes between conditions, and whether the change is attributable to applied mechanical stress rather than altered surface chemistry, DNA-protein interactions, particle survival, or orientational selection.

In an ideal equilibrium model with an externally imposed mechanical bias $U_c(z)$,

$$p(z \mid c) = \frac{\exp[-\beta(G_0(z) + U_c(z))]}{\int \exp[-\beta(G_0(z') + U_c(z'))]\mathrm{d}z'}, \qquad \beta = (k_B T)^{-1}. \tag{47}$$

This is a physically motivated *hypothesis*, not something that a standard cryo-EM classification package automatically infers. Without calibrated $U_c(z)$ and matched controls, observing a structural shift does not by itself determine an applied force.

13.3 Why two conditions can differ even without different conformations

The measured particle distribution is affected by particle inclusion and detection. If state z is selected with probability $s(z, c)$, the distribution in the analyzed dataset is proportional to

$$p_{\text{obs}}(z \mid c) \propto s(z, c)\, p_{\text{true}}(z \mid c). \tag{48}$$

Thus a conformational state that is harder to pick or classify can be depleted in the particle stack, even if its true solution abundance is unchanged. Preferred orientation can also change between experimental conditions. This is why identical processing settings and careful diagnostics help but do not alone remove selection bias.

13.4 A validation ladder for force-dependent structural claims

A defensible analysis should first demonstrate that the common rigid reconstruction and orientation assignment are stable. Next, it should show that structurally different states can be recovered in both independent particle subsets. Then it should compare relaxed, hybridized, and ideally nicked/gapped or otherwise mechanically inactive controls using the *same attachment-site mutant and chemistry*. The recovered state differences should be visible in particle-level likelihoods or reproducible class/latent assignments, not only in a generated animation.

For a learned coordinate, compare class occupancy or coordinate density between conditions only after checking orientation and CTF distributions. If the coordinate will be interpreted as mechanical extension or force, independently calibrate its structural geometry and spring mechanics. In particular, a generic 3DVA or 3DFlex latent coordinate is not a force in piconewtons.

Interpretation warning

The critical causal distinction. A DNA-bearing sample and a DNA-free sample differ in composition, possible mutations, interaction chemistry, and mechanical loading. The difference between their maps can be real but cannot automatically be ascribed to force. A matched mechanical-state comparison is stronger than a comparison between entirely different constructs.

13.5 How should one select an analysis method?

For an initial PP2A study, start with a rigorously validated consensus map and inspect 3DVA. If 3DVA separates a small-amplitude bending direction robustly, this is a useful representation even if it is not an exact mechanical coordinate. If a large trajectory bends back in PCA space, do not derive a Boltzmann free-energy landscape along that folded coordinate. Compare discrete classification, RECOVAR, and nonlinear models, emphasizing reproducibility and uncertainty. For flexible tethered DNA, consider a protein-focused structural analysis alongside a separate investigation of the DNA density; simply masking the DNA may remove nuisance variation *and* potentially important experimental information.

14 Worked statistical problems that connect the theory

14.1 How image averaging improves signal-to-noise

Assume images are aligned perfectly and contain the same underlying signal s plus independent, zero-mean noise: $y_i = s + \epsilon_i$ with $\mathrm{Var}(\epsilon_i) = \sigma^2$. Their mean is

$$\bar{y} = \frac{1}{N}\sum_i y_i = s + \frac{1}{N}\sum_i \epsilon_i.$$

Its noise variance is

$$\mathrm{Var}(\bar{y} - s) = \frac{\sigma^2}{N},$$

so the noise standard deviation decreases as $\sigma/\sqrt{N}$. For fixed signal amplitude, amplitude SNR increases as $\sqrt{N}$.

Worked example

Why ten times the particles does not guarantee ten times the detail. If $N = 100$ aligned observations give an amplitude SNR of 2 at some frequency, ideal independent-noise averaging with $N = 10{,}000$ increases amplitude SNR by $\sqrt{10000/100} = 10$, so the SNR becomes 20. This scaling fails if alignment errors, structural heterogeneity, preferred orientation, radiation damage, or correlated noise dominate.

14.2 Posterior uncertainty and confidence

A maximum-posterior pose estimate can conceal uncertainty. Suppose a particle has two nearly equal pose probabilities, $p(R_A \mid y) = 0.55$ and $p(R_B \mid y) = 0.45$. Reporting only "pose A" discards the fact that 45% posterior probability remains elsewhere. Marginalized reconstruction lets both hypotheses contribute appropriately, while hard assignment can lock the wrong explanation into later iterations. This is especially important at low SNR.

14.3 Population fractions and free energy

Suppose a two-state protein has genuine equilibrium populations $p_A = 0.8$ and $p_B = 0.2$. From Boltzmann statistics,

$$\frac{p_B}{p_A} = \exp[-\beta(F_B - F_A)],$$

so

$$\boxed{F_B - F_A = -k_B T \log \frac{p_B}{p_A} = k_B T \log 4 \approx 1.386 k_B T.} \tag{49}$$

The formula is straightforward. The hard part is establishing that particle classification has recovered unbiased thermodynamic populations. Classification probabilities influenced by sampling, ice effects, unmodeled states, or picking cannot be inserted into this expression without qualification.

14.4 How hard assignments underestimate ambiguity

Suppose three particles have class-A posteriors $(0.9, 0.6, 0.1)$. The soft expected number of A particles is $0.9 + 0.6 + 0.1 = 1.6$, giving an expected fraction $1.6/3 \approx 0.533$. Hard assignment at a 0.5 threshold produces two A particles, fraction $2/3 \approx 0.667$. The difference is not evidence of error; it reflects the uncertainty discarded by hard classification. Always distinguish posterior occupancy estimates from simple counts of assigned labels.

14.5 Why more PCA dimensions may help and hurt

Suppose estimated principal eigenvalues are $(6, 3, 1)$, for a total variance of 10. One component captures 60% of the model variance; two capture 90%. But including additional components can also add noise-dominated directions and make density estimation less reliable, especially when particles provide weak information about small-amplitude motions. The number of meaningful modes should be chosen using stability and predictive evidence, not simply by maximizing the reported explained-variance percentage.

15 Self-test exercises with worked solutions

Exercise 1. Fourier-slice theorem

A 2D object $f(x, z)$ is projected along z, yielding $p(x) = \int f(x, z)\mathrm{d}z$. Explain which slice of $\widehat{f}(k_x, k_z)$ is measured.
Solution. $\widehat{p}(k_x) = \int p(x)e^{-2\pi i k_x x}\mathrm{d}x = \iint f(x, z)e^{-2\pi i(k_x x+0z)}\mathrm{d}x\mathrm{d}z = \widehat{f}(k_x, 0)$. This is the line through the origin at $k_z = 0$.

Exercise 2. Bayes' rule with unequal priors

Two states have priors $\pi_A = 0.8$ and $\pi_B = 0.2$. The measured image is twice as likely under state B as under state A. Find $p(B \mid y)$.
Solution. Set $p(y \mid A) = L$, $p(y \mid B) = 2L$. Bayes gives $p(B \mid y) = 0.2(2L)/[0.8L + 0.2(2L)] = 1/3$. State B has higher likelihood, but the more prevalent prior A still dominates the posterior.

Exercise 3. PCA variance

A covariance matrix is $\Sigma = \left(\begin{smallmatrix} 5 & 0 \\ 0 & 2 \end{smallmatrix}\right)$. Give the first PCA axis and variance explained.
Solution. The eigenvectors are coordinate axes; PC1 is $(1, 0)^T$, and it explains $5/(5 + 2) = 71.4\%$ of the variance.

Exercise 4. Likelihood-based pose comparison

Two candidate poses have squared weighted residuals 18 and 24. Under a Gaussian likelihood with log density $-\frac{1}{2}\times$ weighted residual plus constants, what is the likelihood ratio of the first pose to the second?
Solution. $\exp[-18/2 + 24/2] = e^3 \approx 20.1$. This demonstrates why quite modest per-image fit differences can matter statistically.

Exercise 5. CTF zeros

One image has $C_1 = 0$ at a frequency and another has $C_2 = 0.7$. Which contributes information for reconstruction?
Solution. Only the second image carries first-order phase-contrast information at that frequency. If both are zero, the data provide no such constraint there; a regularizer may stabilize an estimate but cannot restore genuinely unrecorded signal.

Exercise 6. Gold-standard FSC

An FSC curve crosses 0.143 at $q = 0.20\,\text{Å}^{-1}$. State the nominal global resolution and one limitation of the number.
Solution. $d = 1/q = 5.0\,\text{Å}$. Local flexible regions or undersampled directions may have worse resolution.

Exercise 7. Hard versus soft classes

A particle has posterior $p(A \mid y) = 0.51$, $p(B \mid y) = 0.49$. Explain what information is lost when it is hard-assigned to A.
Solution. A hard label says only "A," concealing that B is almost equally plausible. In an EM iteration, retaining the posterior would allow its structural evidence to be weighted approximately 51% for A and 49% for B.

Exercise 8. Linear versus nonlinear motion

A particle varies along a curved manifold. Can a high-dimensional PCA model approximate it, and what remains problematic about a single PCA coordinate?

Solution. Many linear components can approximate a curved trajectory in a high-dimensional space. A single component may not parameterize the path injectively, and PCA geometry alone does not define physical distance, kinetics, or a calibrated reaction coordinate.

Exercise 9. Condition-dependent state populations

A relaxed sample yields 30% particles in class B and a spring-loaded sample yields 60%. What can be claimed immediately, and what cannot?

Solution. Under the fitted classification model, the analyzed stacks have different assigned class fractions. One cannot immediately infer a force, binding free-energy shift, or causal spring effect without considering sampling biases, sample chemistry, pose/CTF differences, model robustness, and a justified mapping to equilibrium populations.

16 Summary: six statements worth retaining

1. **Fourier-slice theorem:** one projection samples a central plane of the three-dimensional Fourier volume. Different orientations are the source of the missing depth information.
2. **Reconstruction is inverse modeling:** solve for a 3D map that predicts the CTF-filtered, shifted, noisy particle images. Regularization prevents unstable inversion.
3. **Alignment is latent-variable inference:** every particle has an unknown rotation and translation, which may need to be marginalized rather than assigned with certainty.
4. **Classification is probabilistic:** discrete states are represented by a mixture of volumes, with posterior responsibilities rather than necessarily certain labels.
5. **PCA is a variance model, not a mechanistic law:** 3DVA and RECOVAR infer structural variability in lower-dimensional representations, but a PCA coordinate need not equal a force, angle, or thermodynamic reaction coordinate.
6. **Nonlinear models add expressive power and new assumptions:** cryoDRGN learns density variation, while 3DFlex learns canonical-density deformation. Neither eliminates the need for independent validation and explicit biological controls.

Notation reference

Symbol	Definition
$V(\boldsymbol{r})$	Three-dimensional reconstruction volume (potential-derived density).
P_R	Projection operator at three-dimensional orientation R.
$R \in SO(3)$	Proper orthogonal 3×3 rotation matrix.
$\boldsymbol{t}_i$	In-plane translation of particle i.
$C_i(\boldsymbol{q})$	Particle-specific contrast transfer function.
y_i, Y_i	Observed particle and its two-dimensional Fourier transform.
ε_i	Random observation noise.
A_i	Linear imaging operator, including projection, CTF, and shift.
Σ_i	Observation-noise covariance for particle i.
π_k	Prior mixing fraction of discrete class k.
γ_{ik}	Posterior responsibility of class k for particle i.
Σ_V	Covariance of three-dimensional molecular volumes.
U_j	PCA / 3D variability difference-volume mode j.
z_i	Continuous latent conformation coordinate for particle i.
$G_\theta(z)$	Learned generative map from latent coordinates to a 3D volume.
$\mathrm{FSC}(q)$	Fourier shell correlation between independent half maps.
$k_B T$	Thermal energy scale; $\beta = (k_B T)^{-1}$.
$\mathcal{F}$	Fourier transform using $e^{-2\pi i \mathbf{q}\cdot\mathbf{r}}$.

Primary literature and software documentation

References

[1] Singer, A. and Sigworth, F. J. (2020). Computational methods for single-particle electron cryomicroscopy. *Annual Review of Biomedical Data Science*, 3, 163–190. doi:10.1146/annurev-biodatasci-021020-093826.

[2] van Heel, M. (1987). Angular reconstitution: a posteriori assignment of projection directions for 3D reconstruction. *Ultramicroscopy*, 21, 111–123. doi:10.1016/0304-3991(87)90078-7.

[3] Scheres, S. H. W. (2012). A Bayesian view on cryo-EM structure determination. *Journal of Molecular Biology*, 415, 406–418. doi:10.1016/j.jmb.2011.11.010.

[4] Scheres, S. H. W. (2012). RELION: implementation of a Bayesian approach to cryo-EM structure determination. *Journal of Structural Biology*, 180, 519–530. doi:10.1016/j.jsb.2012.09.006.

[5] Punjani, A., Rubinstein, J. L., Fleet, D. J. and Brubaker, M. A. (2017). cryoSPARC: algorithms for rapid unsupervised cryo-EM structure determination. *Nature Methods*, 14, 290–296. doi:10.1038/nmeth.4169.

[6] Punjani, A., Zhang, H. and Fleet, D. J. (2020). Non-uniform refinement: adaptive regularization improves single-particle cryo-EM reconstruction. *Nature Methods*, 17, 1214–1221. doi:10.1038/s41592-020-00990-8.

[7] Scheres, S. H. W. and Chen, S. (2012). Prevention of overfitting in cryo-EM structure determination. *Nature Methods*, 9, 853–854. doi:10.1038/nmeth.2115.

[8] Pintilie, G. and Chiu, W. (2021). Validation, analysis and annotation of cryo-EM structures. *Acta Crystallographica D*, 77, 1142–1152. doi:10.1107/S2059798321006069.

[9] Dempster, A. P., Laird, N. M. and Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society B*, 39, 1–22. doi:10.1111/j.2517-6161.1977.tb01600.x.

[10] Tipping, M. E. and Bishop, C. M. (1999). Probabilistic principal component analysis. *Journal of the Royal Statistical Society B*, 61, 611–622. doi:10.1111/1467-9868.00196.

[11] Punjani, A. and Fleet, D. J. (2021). 3D variability analysis: resolving continuous flexibility and discrete heterogeneity from single-particle cryo-EM. *Journal of Structural Biology*, 213, 107702. doi:10.1016/j.jsb.2021.107702.

[12] Gilles, M. A. and Singer, A. (2025). Cryo-EM heterogeneity analysis using regularized covariance estimation and kernel regression. *PNAS*, 122, e2419140122. doi:10.1073/pnas.2419140122.

[13] Zhong, E. D., Bepler, T., Berger, B. and Davis, J. H. (2021). CryoDRGN: reconstruction of heterogeneous cryo-EM structures using neural networks. *Nature Methods*, 18, 176–185. doi:10.1038/s41592-020-01049-4.

[14] Punjani, A. and Fleet, D. J. (2023). 3DFlex: determining structure and motion of flexible proteins from cryo-EM. *Nature Methods*, 20, 860–870. doi:10.1038/s41592-023-01853-8.

[15] CryoSPARC Guide. Ab-Initio Reconstruction: theory, branch-and-bound and SGD. Documentation consulted October 2026.

[16] CryoSPARC Guide. Expectation Maximization in Cryo-EM. Documentation consulted October 2026.

[17] CryoSPARC Guide. Heterogeneous Refinement. Documentation consulted October 2026.

[18] CryoSPARC Guide. 3D Classification tutorial. Documentation consulted October 2026.

[19] CryoSPARC Guide. Homogeneous Refinement. Documentation consulted October 2026.

[20] CryoSPARC Guide. 3D Variability. Documentation consulted October 2026.

[21] CryoSPARC Guide. 3D Flexible Refinement tutorial and limitations. Documentation consulted October 2026.